Jing Huang

Ph.D. Candidate, Educational Psychology & Research Methodology
Purdue University

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Education

Purdue University — Aug. 2023 – Present
Ph.D., Educational Psychology and Research Methodology (Advisor: Hua-Hua Chang) — West Lafayette, USA

Purdue University — Jan. 2026 – Present
M.S., Statistics (concurrent with the Ph.D.) — West Lafayette, USA

East China Normal University — Sep. 2020 – Jun. 2023
M.Ed., Educational Psychology — Shanghai, China

East China Normal University — Sep. 2016 – Jun. 2020
B.S., Statistics — Shanghai, China

Research Interests

I develop methods for adaptive, responsive, and self-renewing live digital assessment under changing conditions:

- **Real-Time Assessment Monitoring** of examinee behavior and item performance throughout live assessment
- **In-Session Intervention** to improve the accuracy and fairness of ability estimation
- **Continuous Item-Pool Maintenance** across administrations through online calibration, content screening, and replenishment with AI-assisted item generation

Honors and Awards

ETS Harold Gulliksen Psychometric Research Fellowship — Aug. 2026
NCME Graduate Student Works-in-Progress eBoard Competition, Award Winner — Apr. 2026
Psychometric Society Vector Travel Award — Jul. 2025
Ross Fellowship, Purdue University — Mar. 2023

Prior to Ph.D.

National Scholarship (top 1%), Ministry of Finance & Ministry of Education, China — Oct. 2022
Excellent Graduate (top 2%), East China Normal University — Oct. 2022
Shu Qing Encouragement Scholarship (top 1%), East China Normal University — Oct. 2021
Third Prize, Contemporary Graduate Statistical Contest in Modeling (Team Leader) — Dec. 2020
Second Prize, COVID-19 Data Analysis Contest (Team Leader) — Jul. 2020
National Encouragement Scholarship (top 2%), East China Normal University — Dec. 2019
Third Prize, Contemporary Undergraduate Statistical Contest in Modeling (Team Leader) — Nov. 2019
Shanghai Popular Science Education Innovation Award, Second Prize — Nov. 2019
Meritorious Winner, Interdisciplinary Contest in Modeling (ICM), COMAP — Feb. 2019
Third-Class Scholarship for Excellent Student, East China Normal University — 2017, 2019

Publications

Summary: 10 peer-reviewed journal articles published (**5 as first author**); 5 manuscripts under review or revision (2 as first author, 1 as co-first author); 2 book chapters; 12 conference presentations (11 as first author).

* *Equal contribution.*

Peer-Reviewed Journal Articles

1. **Huang, J.**, & Chang, H.-H. (2026). Real-Time Monitoring of Item Parameter Drift in CD-CAT via Adaptive CUSUM Charts. *Psychometrika*, 1–61. [doi]

2. **Huang, J.**, & Chang, H.-H. (2026). A Cognitive Design System Approach to Enhancing Cognitive Alignment and Difficulty Control in Gen-AI Item Generation. *Applied Measurement in Education*, 1–21. [doi]

3. Mo, F., **Huang, J.**, Yang, Y., Özen, Z., Maeda, Y., & Olenchak, F. R. (2026). Undergraduate Students' Learning Outcomes with ChatGPT: A Meta-Analytic Study. *Computers and Education: Artificial Intelligence*, 100536. [doi]

4. **Huang, J.**, Zhang, Y., Morphew, J. W., Nissen, J. M., Van Dusen, B., & Chang, H.-H. (2025). Two-phase Content-balancing CD-CAT Online Item Calibration. *Journal of Educational Measurement, 62*(4), 787–808. [doi]

5. **Huang, J.**, Xin, Y. P., & Chang, H.-H. (2025). The Application of Machine Learning to Educational Process Data Analysis: A Systematic Review. *Education Sciences, 15*(7), 888. [doi]

6. Zheng, C., Liu, J., Li, Y., Xu, P., Zhang, B., Wei, R., Liu, B., & **Huang, J.** (2024). A 2PLM-RANK Multidimensional Forced-Choice Model and Its Fast Estimation Algorithm. *Behavior Research Methods*, 1–26. [doi]

7. Yang, X., **Huang, J.**, Chen, X., & Su, X. (2022). A Conceptual Analysis of Conceptual Understanding. *Exploring Education Development, 20*, 54–68. (in Chinese)

8. Li, Q., **Huang, J.**, & Xu, X. (2021). Relationship Between COVID-19 Infection Pneumonia and Traditional Chinese Medicinal Materials. *Chinese Journal of Medical Library and Information Science, 30*(01), 17–24. (in Chinese)

9. **Huang, J.**, & Xu, X. (2020). Research on the Influential Factors of Public Perception and Attitudes to COVID-19: Based on the Citizen Survey of Shanghai. *Library Journal, 39*(12), 98–106. (in Chinese)

10. Xu, X., & **Huang, J.** (2020). Research on Executives' Risk Perception and Behavior Choice to COVID-19. *Library Journal, 39*(12), 117–127. (in Chinese)

Manuscripts Under Review or Revision

1. Chang, H.-H., **Huang, J.**, Zhang, Y., Wang, X., Le, V., Mehrabi, A., Salazar, A., Morphew, J. W., Nissen, J. M., & Van Dusen, B. (revised and resubmitted). How We Know What Students Know: Rethinking College STEM Assessments with CD-CATs. *Psychometrika*.

2. **Huang, J.**, Zhang, J., & Chang, H.-H. (revised and resubmitted). A Dual-Dimensional LLM Framework for Automated Item Incidental Content Similarity Analysis in Large-Scale Assessments. *Applied Measurement in Education*.

3. **Huang, J.**, Guo, X., Richburg, A., Kim, S., Jiang, Y., Dallas, A., & Goodman, J. (under review). Leading Signals in Item-Compromise Monitoring: Measurement, Alignment, and Decision Value. *Applied Psychological Measurement*.

4. Tao, Y.*, **Huang, J.***, & Yang, X. (revised and resubmitted). What Drives Students' Everyday Creativity? *Thinking Skills and Creativity*.

5. Zhang, J., Liang, X., & **Huang, J.** (revised and resubmitted). A Multi-Agent Large Language Model Framework for Automated Q-Matrix Generation. *Psychometrika*.

Book Chapters

1. Zhang, Y., **Huang, J.**, & Chang, H.-H. (2025). Innovative Computerized Test Items. In *Encyclopedia of Social Measurement*. Elsevier. [doi]

2. Wang, S., **Huang, J.**, & Chang, H.-H. (2026). Computerized Adaptive Testing for STEM Learning. In *Artificial Intelligence for STEM Education Research: Advanced Methods and Applications* (pp. 87–111). Cham: Springer Nature Switzerland. [doi]

Works in Progress

1. **Huang, J.**, Fauss, M., Sinharay, S., & Chang, H.-H. (in preparation). Forensic Item Clones: Verifying External AI Assistance and Recovering Ability Estimates in Digital Assessment. *Journal of Educational and Behavioral Statistics*.

2. **Huang, J.**, & Chang, H.-H. Balancing Proficiency and Diagnosis Over Time: A Framework for Dual-Target Adaptive Assessment.

3. **Huang, J.**, & Cheng, M.-M. A Two-Stage Framework Integrating Psychometric Modeling and Process Data for Interpretable DIF Detection.

Conference Presentations

1. **Huang, J.**, & Chang, H.-H. (2026, October). Forensic Item Clones: Verifying External AI Assistance and Recovering Ability Estimates in Digital Assessment. *NCME Artificial Intelligence in Measurement and Education Conference (AIME-Con)*, Pittsburgh, PA, USA.
2. **Huang, J.**, & Chang, H.-H. (2026, April). Adaptive CUSUM Charts for Real-Time Detection of Item Parameter Drift in CD-CAT. *Annual Meeting of the National Council on Measurement in Education (NCME)*, Los Angeles, CA, USA.
3. Wang, T., Yang, X., & **Huang, J.** (2026, April). A Mixture CDM for Mathematical Problem-Solving Strategies. *Annual Meeting of the National Council on Measurement in Education (NCME)*, Los Angeles, CA, USA.
4. **Huang, J.**, Zhang, Y., Wang, X., & Chang, H.-H. (2026, April). Optimizing Cognitive Diagnosis CAT: Addressing Practical Challenges through Psychometric Innovation. *Annual Meeting of the American Educational Research Association (AERA)*, Los Angeles, CA, USA.
5. **Huang, J.**, Zhang, J., & Chang, H.-H. (2026, April). LLM-Enhanced Automated Item Content Analysis Supporting CAT Applications. *Annual Meeting of the American Educational Research Association (AERA)*, Los Angeles, CA, USA.
6. **Huang, J.**, & Chang, H.-H. (2025, November). Generative AI-Enhanced Cognitive Design System Approach to Automatic Item Generation. *Purdue AI in P-12 Education Conference*, West Lafayette, IN, USA.
7. **Huang, J.**, & Chang, H.-H. (2025, July). Bridging Psychometrics and Generative AI: Automatic Item Generation for Verbal Analogical Reasoning. *International Meeting of the Psychometric Society (90th Annual Meeting)*, Minneapolis, MN, USA.
8. **Huang, J.**, Zhang, Y., Morphew, J. W., Nissen, J. M., Van Dusen, B., & Chang, H.-H. (2025, April). CD-CAT Online Item Calibration with Practical Constraints. *Annual Meeting of the National Council on Measurement in Education (NCME)*, Denver, CO, USA.
9. **Huang, J.**, Zhang, Y., & Chang, H.-H. (2024, November). Two-phase Content-balancing CD-CAT Online Item Calibration. *Ideas in Testing Research*, Chicago, IL, USA.
10. **Huang, J.**, Xin, Y. P., & Chang, H.-H. (2024, November). Machine Learning to Educational Process Data Analysis: A Systematic Review. *Purdue AI in P-12 Education Conference*, West Lafayette, IN, USA.
11. **Huang, J.**, Yang, T., & Yang, X. (2024, April). Taking a "Whole" Perspective: Exploring the Influence of Individual and Environmental Factors on Creative Performance. Roundtable, *Annual Meeting of the American Educational Research Association (AERA)*, Philadelphia, PA, USA.
12. **Huang, J.**, & Yang, X. (2022, April). Model-based Approaches to Classifying Students into Competence-based Proficiency Levels. Poster, *Annual Meeting of the American Educational Research Association (AERA)*, San Diego, CA, USA.

Research Experience

Harold Gulliksen Psychometric Research Fellow — ETS Aug. 2026 – Present
Mentors: Sandip Sinharay and Michael Fauss Remote, USA

- Develop sequential methods for real-time detection of aberrant examinee behavior using joint response and response-time data (dissertation research).
- First-authored manuscript on forensic item clones for verifying external AI assistance and recovering ability estimates, in preparation for *Journal of Educational and Behavioral Statistics*.

Psychometric Research Intern — NCCPA Jun. 2026 – Aug. 2026
National Commission on Certification of Physician Assistants Johns Creek, GA, USA

- Developed an LLM-based content-risk scoring rubric for certification items, rated by multiple large language models and evaluated with a generalizability study.
- Built a Bayesian change-point model that uses content-risk scores as prior information to detect compromised items in a longitudinal assessment program.
- First-authored manuscript under review at *Applied Psychological Measurement*.

Graduate Research Assistant — Purdue University Aug. 2024 – Jul. 2025
School of Engineering Education; NSF Grant No. 2141847 West Lafayette, USA

- Developed, maintained, and refined CD-CAT algorithms; led parameter estimation, item-selection optimization, online calibration, and CAT tuning procedures; conducted simulation studies.
- Supported mathematics cognitive-diagnosis development and psychometric refinement; performed psychometric analyses of operational LASSO-platform data.

Consultant — Psychometric Investigation Laboratory Dec. 2023 – Present
Purdue University West Lafayette, USA

- Advise on statistical methods, research design, and data management.

Research Assistant — East China Normal University Apr. 2021 – Jun. 2023
Suzhou Micro-Monitoring Project; "13th Five-Year Plan" Project Shanghai, China

- Modeled creativity data from 4,833 students across 48 schools using Rasch analysis and multilevel models.
- Analyzed teacher–student interactions from classroom video data.

Research Assistant — Chinese Hospital Association & ECNU Jul. 2021 – Sep. 2021
Clinical Pharmacist Training Evaluation Shanghai, China

- Analyzed student-satisfaction factors using ordinal logistic regression.

Research Assistant — Surveys and Data Center, ECNU Sep. 2019 – Jun. 2020
Quality-of-Life Survey Studies Shanghai, China

- Analyzed and cleaned large-scale survey data; co-authored two journal papers.

Teaching

Graduate Teaching Assistant — Purdue University Aug. 2025 – May 2026
College of Education West Lafayette, USA

- EDPS 32700 Classroom Assessment (Dr. Anne Traynor).

Invited Guest Lecturer — Purdue University Apr. 2026
College of Education West Lafayette, USA

- EDPS 63600 Item Response Theory: led an R tutorial on IRT parameter estimation with the `mirt` package and an anchor-test linking exercise.

Workshop Instructor — Psychometric Investigation Laboratory Dec. 2023 – Present
Purdue University West Lafayette, USA

- *Using Machine Learning to Analyze Educational Data* workshop series.

Professional Service

Conference Reviewer

NCME Annual Meeting — Individual Paper Proposals (10 proposals) 2027
AERA Annual Meeting — SIG-Computer and Internet Applications in Education (3 proposals) 2027
NCME AIME-Con — Innovation Demonstration & Paper/Poster Session Review Modules (7 proposals) 2026

Journal Reviewer

Psychometrika 2025, 2026
Journal of Computerized Adaptive Testing 2025